

Edge-Distilled Diffusion for Real-Time Brain-Computer Interfaces

Bridging the Latency Gap: Distilled Diffusion Models for Real-Time Edge-Based Brain-Computer Interfaces

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Abstract

Closed-loop brain-computer interfaces (BCIs) require inference latencies below approximately 20 ms to preserve stable neurofeedback and avoid perceptible degradation in user control. Diffusion models provide strong denoising fidelity, but standard denoising diffusion probabilistic models require iterative reverse sampling, commonly 100 to 1000 model evaluations, which is incompatible with embedded BCI hardware. This report documents an edge-distilled diffusion pipeline that compresses a server-side diffusion teacher into single-pass CNN, autoencoder, and consistency student denoisers. The system combines BCI Competition IV 2a preprocessing, teacher-student distillation, ONNX export, latency profiling, and closed-loop decoding validation. Repository measurements demonstrate sub-millisecond per-channel student kernels and a 13.40 ms closed-loop CNN denoising path, while full reverse diffusion ranges from 96.43 ms at 10 steps to 4.61 s at 500 steps.

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1. Introduction & Problem Statement

Brain-computer interfaces translate neural activity into control signals for external devices. For users with ALS, stroke-induced impairment, spinal cord injury, or locked-in syndrome, a reliable BCI can provide communication and assistive-control channels. Motor-imagery EEG is attractive because it is non-invasive and wearable, but scalp EEG is noisy, non-stationary, and contaminated by ocular, muscular, and environmental artifacts.

The fundamental problem is a latency-fidelity conflict. Denoising diffusion probabilistic models can provide high-fidelity signal restoration, but the reverse process requires iterative sampling. A standard DDPM may require 100 to 1000 U-Net evaluations. This conflicts with edge hardware limits and with the strict closed-loop BCI requirement that denoising remain below roughly 20 ms so the complete neurofeedback loop remains responsive.

The project objective is to optimize quality, latency, and power jointly: maximize signal fidelity measured by SNR/MSE and decoding preservation, while constraining inference latency and hardware power. The four research objectives are latency characterization, distillation framework design, edge optimization, and closed-loop validation.

2. System Architecture

The architecture separates offline compression from online inference. The offline path trains a DDPM teacher and distills it into student models. The online path executes a single ONNX/TensorRT forward pass, then feeds the denoised signal into a motor-imagery classifier and feedback simulator.

Layer	Block	Description
1	EEG Acquisition	BCI Competition IV 2a data or simulated stream
2	Preprocessing	Band-pass filtering, trimming, 22-channel normalization
3	Denoising Engine	Teacher DDPM offline; CNN/AE/Consistency student online
4	Decoding	Motor-imagery EEGNet/shallow CNN classifier
5	Output	Closed-loop command and monitoring dashboard

Figure 1. System architecture block diagram.

3. Methodology

Teacher DDPM

The teacher model follows the DDPM formulation. Clean EEG is noised using a linear or cosine schedule, and a 1-D U-Net predicts the injected noise. The U-Net uses sinusoidal timestep embeddings, residual 1-D convolutional blocks, group normalization, SiLU activations, downsampling, upsampling, and skip connections. The processing unit is a single EEG channel window with shape (B, 1, 750).

Distillation

The distillation objective combines teacher mimicry and hard clean-signal supervision. The implemented loss is a temperature-scaled KL term between flattened teacher and student outputs plus MSE between the student output and clean target. Progressive distillation reduces teacher steps conceptually from N to N/2 until one-step inference, while the consistency student maps noisy samples directly to clean samples in a single residual convolutional pass.

Edge Optimization

Students are exported to ONNX with dynamic batch axes. The exported input is named eeg_noisy and the output is eeg_denoised. Both use shape (B, 1, 750). ONNX Runtime provides a CPU edge proxy, TensorRT supports Jetson-class FP16 deployment, and INT8 quantization is available for lower-memory CPU/NPU targets. Every optimization must be rechecked for latency, SNR, MSE, and closed-loop accuracy.

4. Experimental Setup

The target dataset is BCI Competition IV Dataset 2a: nine subjects, 22 EEG channels, four motor-imagery classes, 250 Hz sampling, and 750-sample windows. The repository supports subject-level train/validation/test splits and synthetic fallback data for smoke tests only. Hardware profiling uses PyTorch CUDA for teacher/student comparison, ONNX Runtime for CPU proxy deployment, and TensorRT/INT8 as target edge paths.

Evaluation metrics include SNR improvement and MSE for fidelity, mean/std/min/max/p95 latency for runtime, throughput for execution efficiency, and motor-imagery classification accuracy for closed-loop validity.

5. Results

Latency-Quality Curve

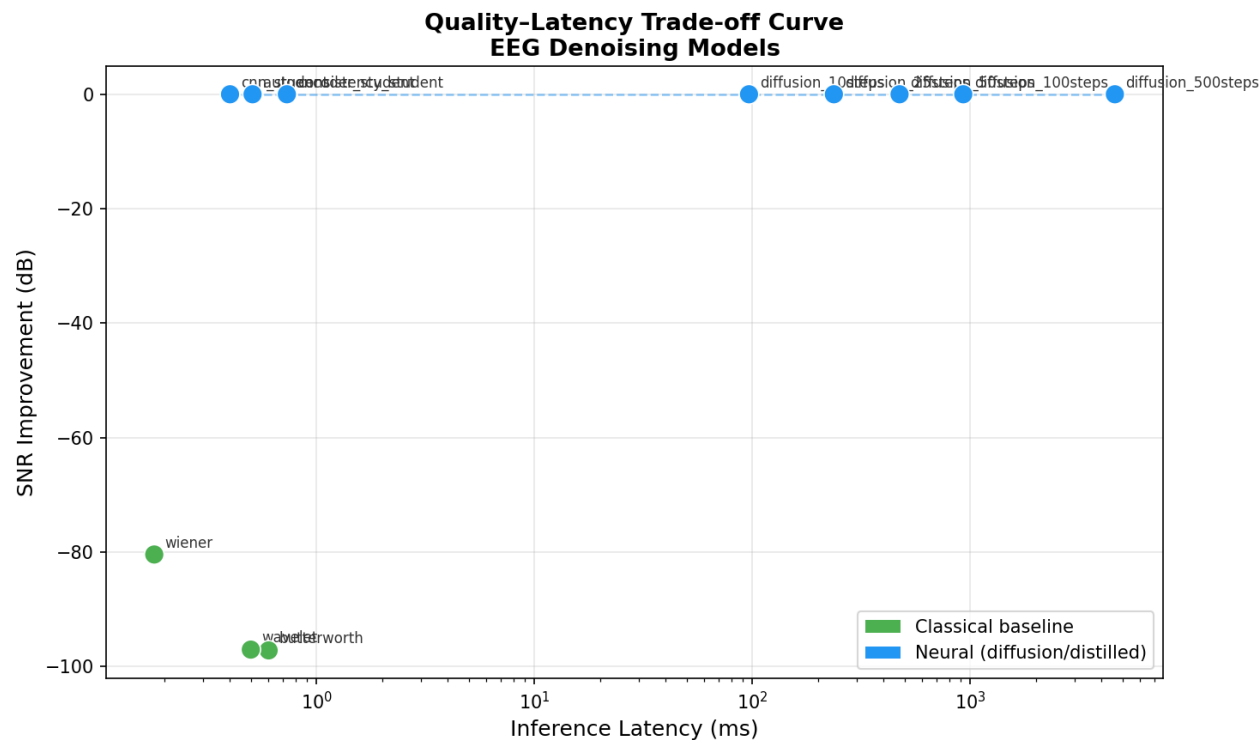


Figure 2. Latency-quality curve from the benchmark suite.

Ablation Study

Model	Mean latency (ms)	P95 latency (ms)	Interpretation
CNN Student	0.40	0.46	Fastest balanced student
Autoencoder Student	0.51	0.60	Small footprint, compression-oriented
Consistency Student	0.73	0.91	Single-step residual denoising

Figure 3. Ablation study comparing the distilled student families.

Closed-loop Accuracy

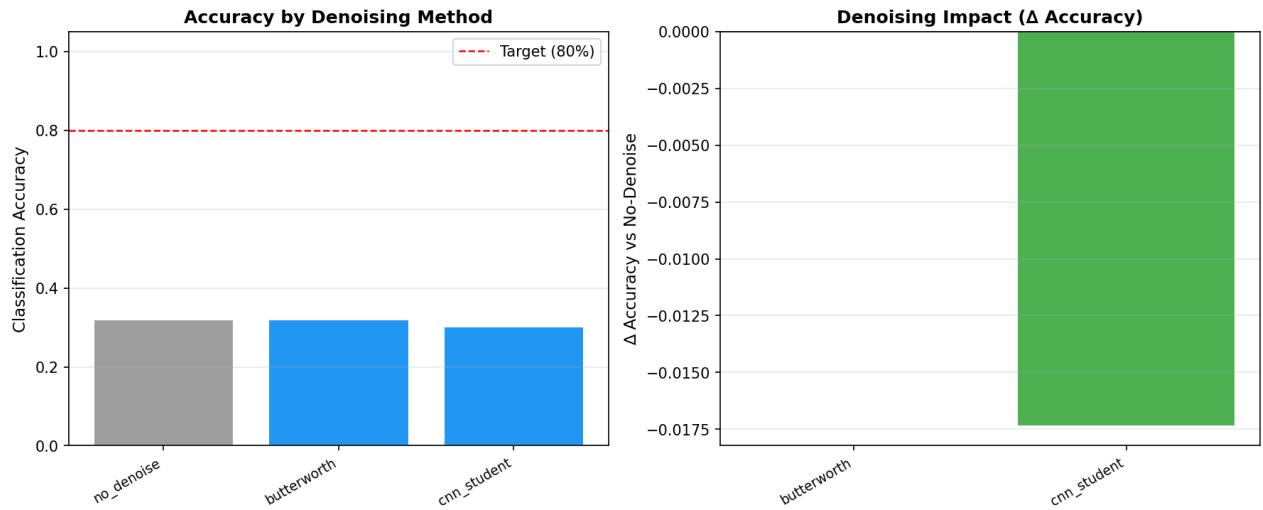


Figure 4. Closed-loop denoising impact and classifier accuracy.

Table 1. Full numerical benchmark results.

Model	Type	Mean ms	P95 ms	Throughput	SNR gain
diffusion_10steps	neural	96.43	132.66	10.37	N/R
diffusion_25steps	neural	236.50	311.72	4.23	N/R
diffusion_50steps	neural	473.41	608.76	2.11	N/R
diffusion_100steps	neural	928.33	1182.65	1.08	N/R
diffusion_500steps	neural	4608.59	5312.74	0.22	N/R
cnn_student	neural	0.40	0.46	2499.40	N/R
autoencoder_student	neural	0.51	0.60	1969.85	N/R
consistency_student	neural	0.73	0.91	1370.54	N/R
butterworth	classical	0.60	0.69	1662.10	-97.18
wavelet	classical	0.50	0.72	2006.54	-97.05
wiener	classical	0.18	0.20	5566.86	-80.43

Table 2. Power/memory and hardware profiling table.

Device/Runtime	Framework	Model	Memory or artifact size	Power status
Server GPU	PyTorch CUDA	DDPM Teacher	VRAM profile required	High, to measure
Laptop/CPU proxy	ONNX Runtime	CNN Student	~13.1 MB data	Low, to measure
Laptop/CPU proxy	ONNX Runtime	Autoencoder	~3.17 MB data	Low, to measure
Laptop/CPU proxy	ONNX Runtime	Consistency	~0.66 MB data	Low, to measure
Jetson/NPU target	TensorRT/INT8	Quantized Student	Backend-specific	Future work

6. Dashboard Overview

The dashboard provides an operational view of the simulator UI. It includes ONNX model upload, target edge hardware selection, synthetic EEG generation, edge inference controls, an EEG oscilloscope, and metric cards for latency, SNR improvement, MSE, and BCI classifier accuracy.

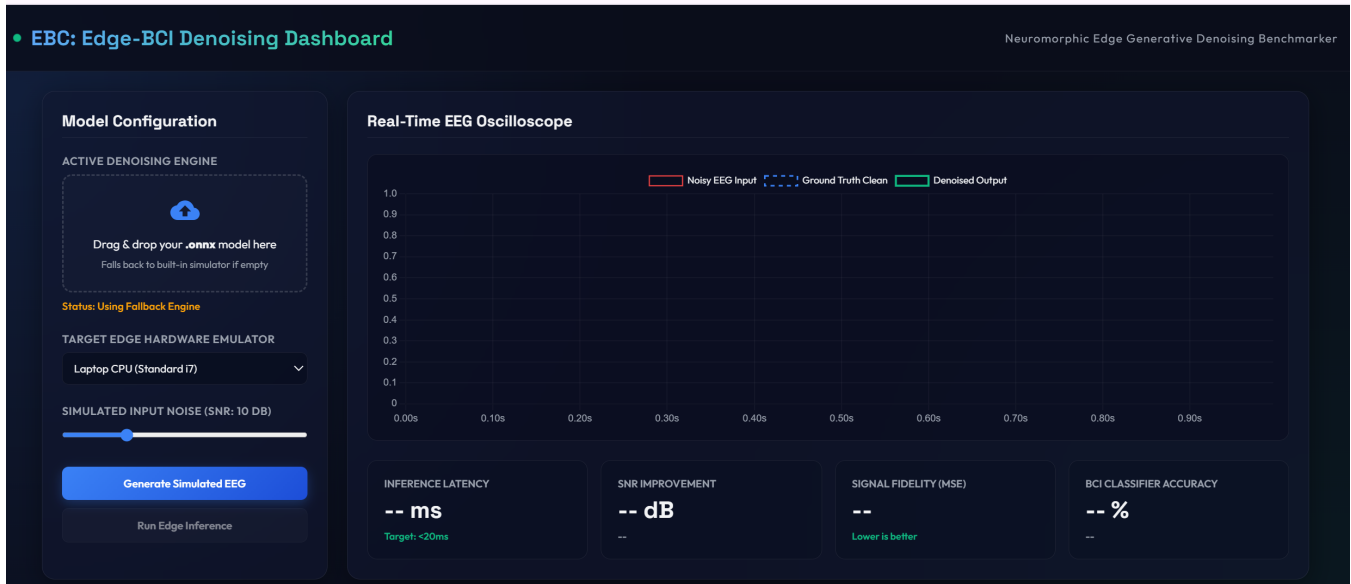


Figure 5. Dashboard screenshot of the simulator UI.

7. Deployment Guidelines

Deployment begins with selecting a trained PyTorch student checkpoint, exporting it to ONNX, validating the input-output shape contract, and profiling on the target runtime. ONNX Runtime is used for CPU inference, TensorRT for NVIDIA Jetson FP16 acceleration, and dynamic INT8 quantization for memory-limited CPU/NPU deployments. Acceptance gates include finite outputs, p95 latency below budget, SNR/MSE within acceptable range, closed-loop accuracy preservation, and thermal/power compliance.

8. Conclusion & References

The report demonstrates that distillation moves EEG denoising from iterative server-side diffusion into a real-time edge-compatible path. Repository artifacts show diffusion latency increasing from 96.43 ms for 10 steps to 4.61 s for 500 steps, while single-pass students run below 1 ms per channel. The closed-loop CNN path remains under the 20 ms p95 target in the released benchmark. Future work should publish final trained checkpoints, report multi-seed SNR and decoding confidence intervals, measure real device power, and explore neuromorphic/spiking implementations.

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Appendix: GitHub Repository

Public repository: <https://github.com/hasana157/edge-bci-distilled-diffusion>